To be sure, this result represented a mere preliminary to the tackling of the real problem, which now loomed all the more formidably before me. My first attempt to overcome it was unsuccessful, for my original silent hope that the radiation emitted by the oscillator would differ, in some characteristic way, from the absorbed radiation, turned out to have been mere wishful thinking. The oscillator reacts only to those rays which it is capable of emitting, and is completely insensitive to adjacent spectral regions.

Moreover, my suggestion that the oscillator was capable of exerting a unilateral, in other words irreversible, effect on the energy of the surrounding field, drew a vigorous protest from Boltzmann, who, with his wider experience in this domain, demonstrated that according to the laws of classical dynamics, each of the processes I considered could also take place in the opposite direction; and indeed in such a manner, that a spherical wave emitted by an oscillator could reverse its direction of motion, contract progressively until it reached the oscillator and be reabsorbed by the latter, so that the oscillator could then again emit the previously absorbed energy in the same direction from which the energy had been received. To be sure, I could exclude such odd phenomena as inwardly directed spherical waves, by the introduction of a specific stipulation—the hypothesis of a natural radiation, which plays the same part in the theory of radiation as the hypothesis of molecular disorder in the kinetic theory of gases, in that it guarantees the irreversibility of the radiation processes. But the calculations showed ever more clearly that an essential link was still missing, without which the attack on the core of the entire problem could not be undertaken successfully.

So I had no other alternative than to tackle the problem once again—this time from the opposite side, namely, from the side of thermodynamics, my own home territory where I felt myself to be on safer ground. In fact, my previous studies of the Second Law of Thermodynamics came to stand me in good stead now, for at the very outset I hit upon the idea of correlating not the temperature but the entropy of the oscillator with its energy. It was an odd jest of fate that a circumstance which on former occasions I had found unpleasant, namely, the lack of interest of my colleagues in the direction taken by my investigations, now turned out to be an outright boon. While a host of outstanding physicists worked on the problem of spectral energy distribution, both from the experimental and theoretical aspect, every one of them directed his efforts solely toward exhibiting the dependence of the intensity of radiation on the temperature. On the other hand, I suspected that the fundamental connection lies in the dependence of entropy upon energy. As the significance of the concept of entropy had not yet come to be fully appreciated, nobody paid any attention to the method adopted by me, and I could work out my calculations completely at my leisure, with absolute thoroughness, without fear of interference or competition.